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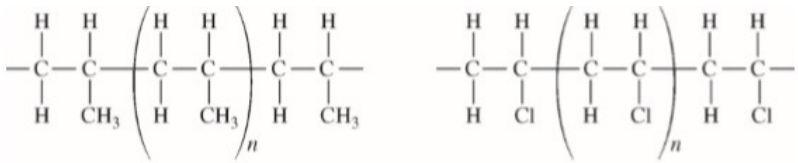
1. A student places a mixture of plastic beads consisting of polypropylene (PP) and polyvinyl chloride (PVC) in a 1.0 L beaker containing distilled water. After stirring the contents of the beaker vigorously, the student observes that the beads of one type of plastic sink to the bottom of the beaker and the beads of the other type of plastic float on the water. The chemical structures of PP and PVC are represented by the diagrams below, which show segments of each polymer.



PP PVC

a. Given that the spacing between polymer chains in PP and PVC is similar, the beads that sink are made of which polymer ? Explain.

PP is synthesized from propene, C_3H_6, and PVC is synthesized from vinyl chloride, C_2H_3Cl. The structures of the molecules are shown below.

Propene Vinyl Chloride (chloroethene)

b. The boiling point of liquid propene (226 K) is lower than the boiling point of liquid vinyl chloride (260 K). Account for this difference in terms of the types and strengths of intermolecular forces present in each liquid.

In a separate experiment, the student measures the enthalpies of combustion of propene and vinyl chloride. The student determines that the combustion of 2.00 mol of vinyl chloride releases 2300 kJ of energy, according to the equation below.

$$2\ C_2H_3Cl(g) + 5\ O_2(g) \rightarrow 4\ CO_2(g) + 2\ H_2O(g) + 2\ HCl(g)\ \Delta H^\circ = -2300\ \text{kJ/mol}_{rxn}$$

c. Using the table of standard enthalpies of formation below, determine whether the combustion of 2.00 mol of propene releases more, less, or the same amount of energy that 2.00 mol of vinyl chloride releases. Justify your answer with a calculation. The balanced equation for the combustion of 2.00 mol of propene is $2\ C_3H_6(g) + 9\ O_2(g) \rightarrow 6\ CO_2(g) + 6\ H_2O(g)$.

Substance	$C_2H_3Cl(g)$	$C_3H_6(g)$	$CO_2(g)$	$H_2O(g)$	$HCl(g)$	$O_2(g)$
Standard Enthalpy of Formation (kJ/mol)	37	21	−394	−242	−92	0

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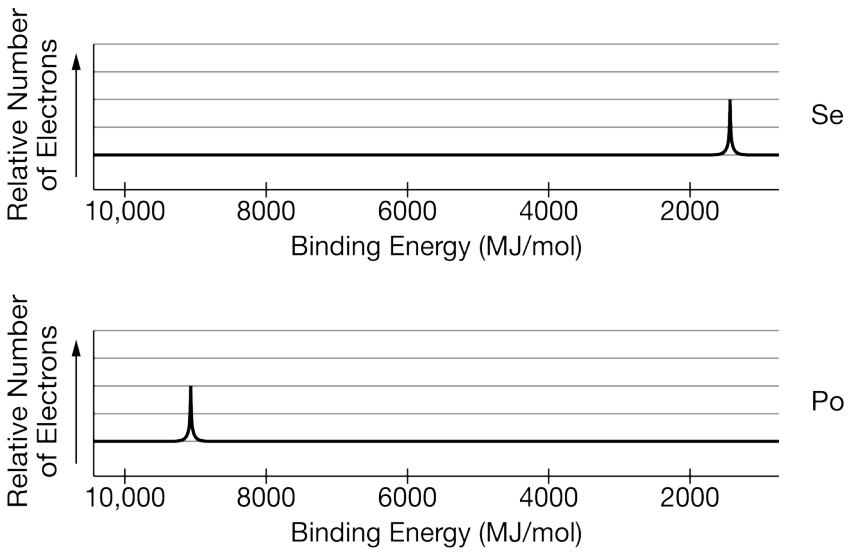
2. **Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

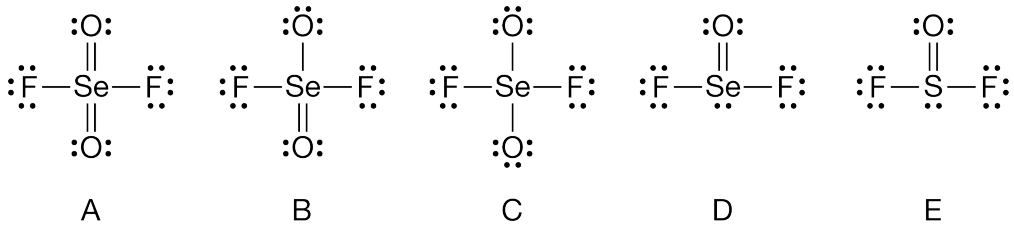
Answer the following questions about elements in Group 16 of the periodic table.

(a) Using principles of atomic structure, explain why the first ionization energy of Po is less than that of Se.

(b) Photoelectron spectroscopy data for the $1s$ sublevel of Se and the $1s$ sublevel of Po are represented below. In terms of Coulomb's law and atomic structure, explain why the peak for Po is positioned so far to the left of the peak for Se.



Three compounds that contain elements from Group 16 are $\mathrm{SeO_2F_2}$, $\mathrm{SeOF_2}$, and $\mathrm{SOF_2}$. Possible Lewis electron dot diagrams for the compounds are shown below.



(c) Which structure, A, B, or C, minimizes the formal charges for each atom in $\mathrm{SeO_2F_2}$? Justify your choice.

(d) A student proposes that the energy of the $\mathrm{Se-to-O}$ bond in the structure in diagram D is greater than that of the $\mathrm{Se-to-O}$ bond in the structure in diagram C. Do you agree or disagree with the student's claim? Justify your answer.

(e) The dipole moment of the $\mathrm{SeO_2F_2}$ molecule (diagrams A, B and C) is nonzero. Identify its molecular geometry and explain how its geometry and the polarity of its bonds result in a nonzero dipole moment.

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The melting point of $SeOF_2$ (diagram D) is greater than the melting point of SOF_2 (diagram E).

(f) Identify the type(s) of intermolecular force(s) that the two substances have in common.

(g) Explain the difference in melting points based on the types and relative strengths of intermolecular forces.

(h) $SeOF_2$ reacts with SiO_2 to produce SeO_2 and SiF_4. The unbalanced chemical equation for the reaction appears below.

$$a\ SeOF_2 + b\ SiO_2 \rightarrow c\ SeO_2 + d\ SiF_4$$

Give the value of each of the following coefficients in the balanced equation.

$a =$
$b =$
$c =$
$d =$

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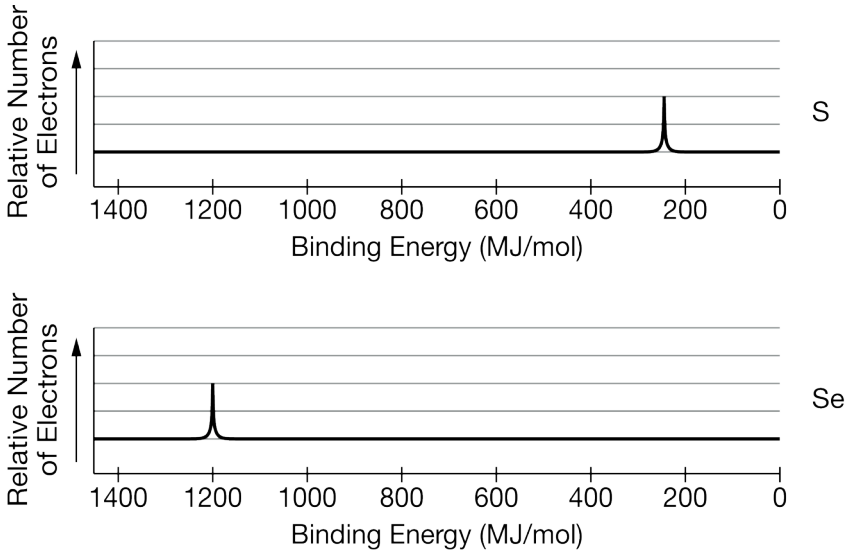
3. **Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

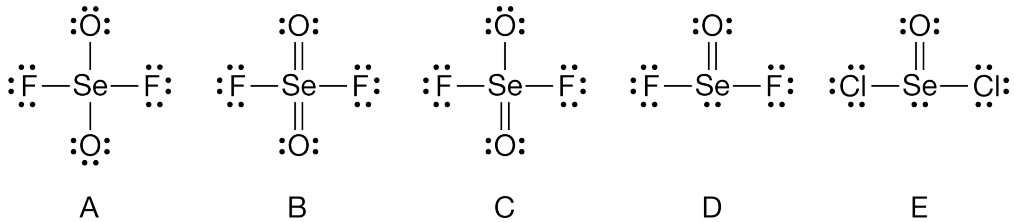
Answer the following questions about elements in Group 16 of the periodic table.

(a) Using principles of atomic structure, explain why the atomic radius of Kr is less than the ionic radius of Se^{2-}.

(b) Photoelectron spectroscopy data for the $1s$ sublevel of S and the $1s$ sublevel of Se are represented below. In terms of Coulomb's law and atomic structure, explain why the peak for Se is positioned so far to the left of the peak for S.



Three compounds that contain elements from Group 16 are $\mathrm{SeO_2F_2}$, $\mathrm{SeOF_2}$, and $\mathrm{SeOCl_2}$. Possible Lewis electron dot diagrams for the compounds are shown below.



(c) Which structure, A, B, or C, minimizes the formal charges for each atom in $\mathrm{SeO_2F_2}$? Justify your choice.

(d) A student proposes that the energy of the $\mathrm{Se-to-O}$ bond in the structure in diagram D is less than that of the $\mathrm{Se-to-O}$ bond in the structure in diagram A. Do you agree or disagree with the student's claim? Justify your answer.

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(e) The dipole moment of the $SeOF_2$ molecule (diagram D) is nonzero. Identify its molecular geometry and explain how its geometry and the polarity of its bonds result in a nonzero dipole moment.

The boiling point of $SeOF_2$ (diagram D) is less than the boiling point of $SeOCl_2$ (diagram E).

(f) Identify the type(s) of intermolecular force(s) that the two substances have in common.

(g) Explain the difference in boiling points based on the types and relative strengths of intermolecular forces.

(h) $SeOF_2$ reacts with $NaOH$ to produce Na_2SeO_3, NaF, and H_2O. The unbalanced chemical equation for the reaction appears below.

$$a\ SeOF_2 + b\ NaOH \rightarrow c\ Na_2SeO_3 + d\ NaF + e\ H_2O$$

Give the value of each of the following coefficients in the balanced equation.

$a =$
$b =$
$c =$
$d =$
$e =$

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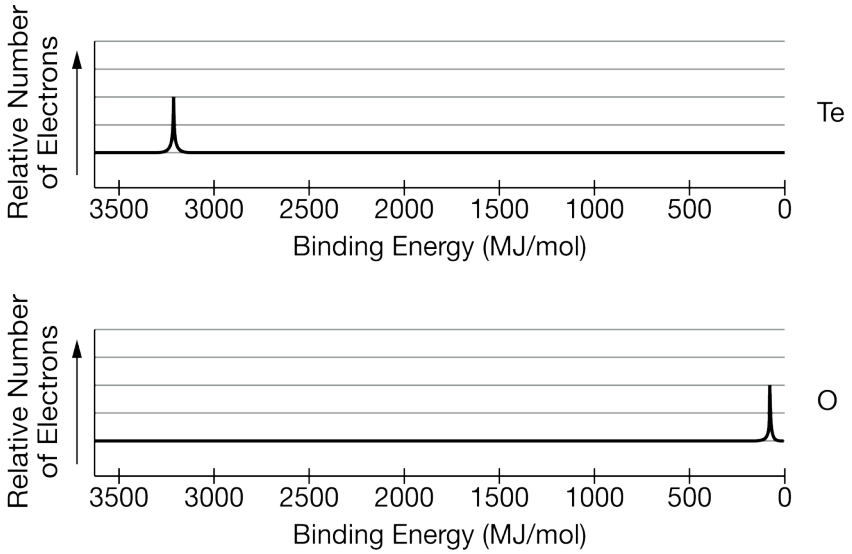
4. **Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate**.

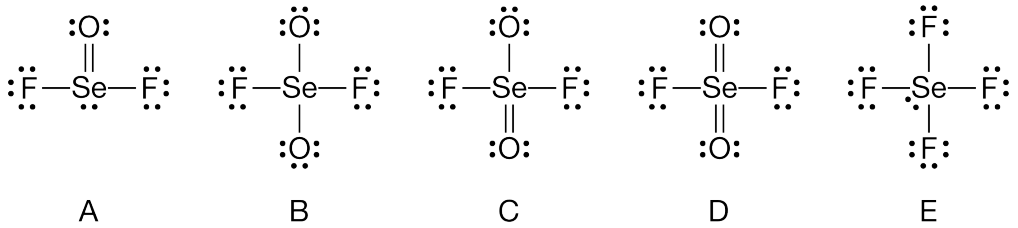
Answer the following questions about elements in Group 16 of the periodic table.

(a) Using principles of atomic structure, explain why the atomic radius of Te is less than the ionic radius of Te^{2-}.

(b) Photoelectron spectroscopy data for the $1s$ sublevel of Te and the $1s$ sublevel of O are represented below. In terms of Coulomb's law and atomic structure, explain why the peak for O is positioned so far to the right of the peak for Te.



Three compounds that contain elements from Group 16 are $\mathrm{SeOF_2}$, $\mathrm{SeO_2F_2}$, and $\mathrm{SeF_4}$. Possible Lewis electron dot diagrams for the compounds are shown below.



(c) Which structure, B, C or D, minimizes the formal charges for each atom in $\mathrm{SeO_2F_2}$? Justify your choice.

(d) A student proposes that the energy of the $\mathrm{Se-to-F}$ bond in the structure in diagram A is equal to that of the $\mathrm{Se-to-F}$ bond in the structure in diagram D. Do you agree or disagree with the student's claim? Justify your answer.

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(e) The dipole moment of the SeF_4 molecule (diagram E) is nonzero. Identify its molecular geometry and explain how its geometry and the polarity of its bonds result in a nonzero dipole moment.

The melting point of $SeOF_2$ (diagram A) is greater than the melting point of SeO_2F_2 (diagrams B, C, and D).

(f) Identify the type(s) of intermolecular force(s) that the two substances have in common.

(g) Explain the difference in melting points based on the types and relative strengths of intermolecular forces.

(h) TeO_2 reacts with SeF_4 to produce $SeOF_2$ and TeF_4. The unbalanced chemical equation for the reaction appears below.

$a\ TeO_2 + b\ SeF_4 \rightarrow c\ SeOF_2 + d\ TeF_4$

Give the value of each of the following coefficients in the balanced equation.

$a =$
$b =$
$c =$
$d =$

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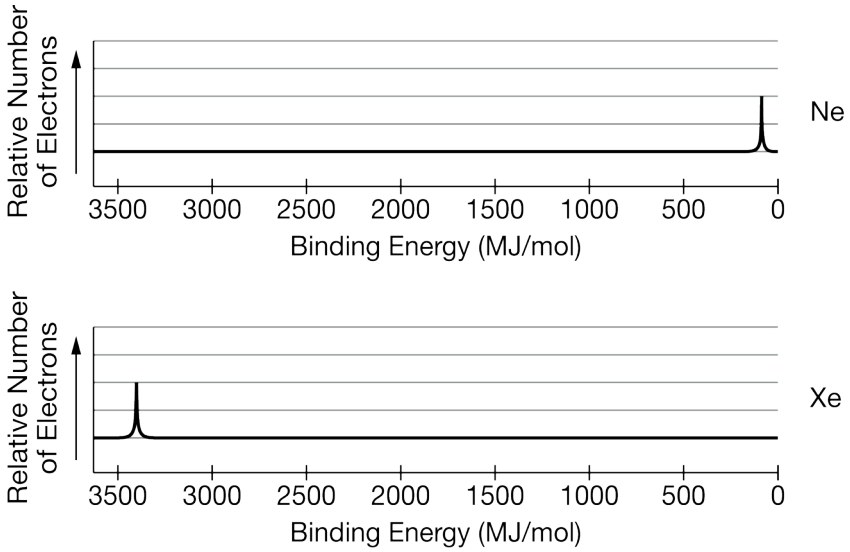
5. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

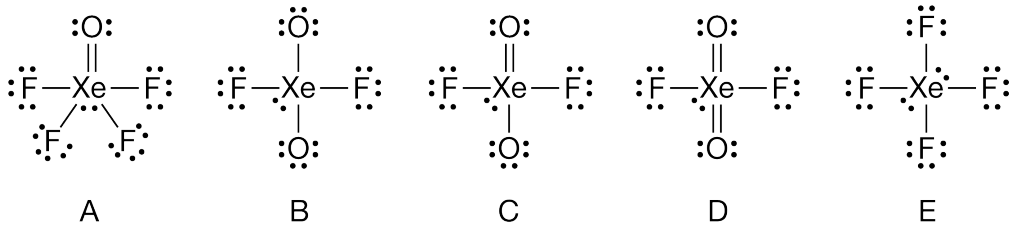
Answer the following questions about elements in Group 18 of the periodic table.

(a) Using principles of atomic structure, explain why the atomic radius of Xe is less than that of Te.

(b) Photoelectron spectroscopy data for the $1s$ sublevel of Ne and the $1s$ sublevel of Xe are represented below. In terms of Coulomb's law and atomic structure, explain why the peak for Ne is positioned so far to the right of the peak for Xe.



Three compounds that contain an element from Group 18 are $\mathrm{XeOF_4}$, $\mathrm{XeO_2F_2}$, and $\mathrm{XeF_4}$. Possible Lewis electron dot diagrams for the compounds are shown below.



(c) Which structure, B, C, or D, minimizes the formal charges for each atom in $\mathrm{XeO_2F_2}$? Justify your choice.

(d) A student proposes that the energy of the $\mathrm{Xe-to-F}$ bond in the structure in diagram B is equal to that of the $\mathrm{Xe-to-F}$ bond in the structure in diagram A. Do you agree or disagree with the student's claim? Justify your answer.

(e) The dipole moment of the $\mathrm{XeOF_4}$ molecule (diagram A) is nonzero. Identify its molecular geometry and

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explain how its geometry and the polarity of its bonds result in a nonzero dipole moment.

The boiling point of XeF_4 (diagram E) is greater than the boiling point of elemental Xe.

(f) Identify the type(s) of intermolecular force(s) that the two substances have in common.

(g) Explain the difference in boiling points based on the types and relative strengths of intermolecular forces.

(h) $XeOF_4$ reacts with SiO_2 to produce XeO_2F_2 and SiF_4. The unbalanced chemical equation for the reaction appears below.

$$a\ SiO_2 + b\ XeOF_4 \rightarrow c\ SiF_4 + d\ XeO_2F_2$$

Give the value of each of the following coefficients in the balanced equation.

$a =$
$b =$
$c =$
$d =$

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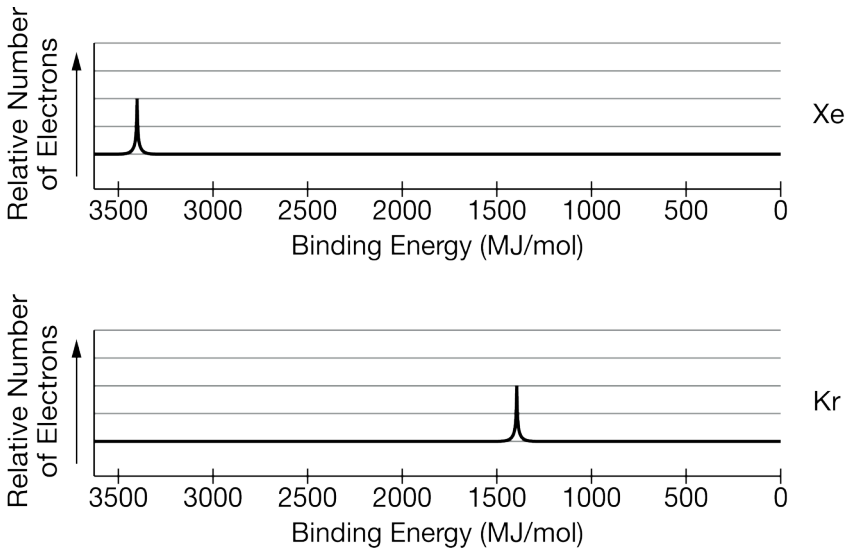
6. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

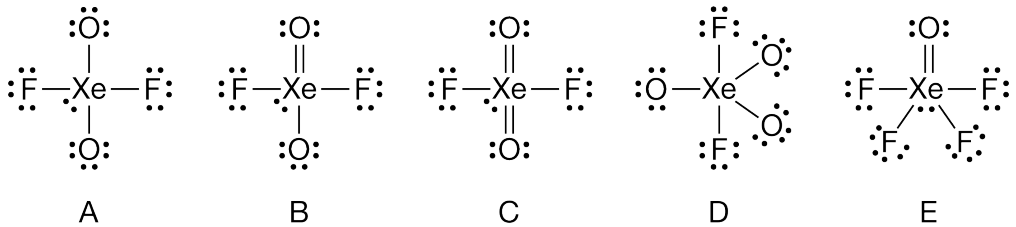
Answer the following questions about elements in Group 18 of the periodic table.

(a) Using principles of atomic structure, explain why the ionic radius of Cs^+ is less than the atomic radius of Xe.

(b) Photoelectron spectroscopy data for the $1s$ sublevel of Xe and the $1s$ sublevel of Kr are represented below. In terms of Coulomb's law and atomic structure, explain why the peak for Kr is positioned so far to the right of the peak for Xe.



Three compounds that contain an element from Group 18 are XeO_2F_2, XeO_3F_2, and $XeOF_4$. Possible Lewis electron dot diagrams for the compounds are shown below.



(c) Which structure, A, B, or C, minimizes the formal charges for each atom in XeO_2F_2? Justify your choice.

(d) A student proposes that the energy of the $Xe - to - O$ bond in the structure in diagram E is greater than that of the $Xe - to - O$ bond in the structure in diagram D. Do you agree or disagree with the student's claim? Justify your answer.

(e) The dipole moment of the XeO_3F_2 molecule (diagram D) is zero. Identify its molecular geometry and explain how its geometry and the polarity of its bonds result in a zero dipole moment.

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The melting point of $XeOF_4$ (diagram E) is greater than the melting point of XeO_3F_2 (diagram D).

(f) Identify the type(s) of intermolecular force(s) that the two substances have in common.

(g) Explain the difference in melting points based on the types and relative strengths of intermolecular forces.

(h) XeF_6 reacts with SiO_2 to produce $XeOF_4$ and SiF_4. The unbalanced chemical equation for the reaction appears below.

$$a\ XeF_6 + b\ SiO_2 \rightarrow c\ XeOF_4 + d\ SiF_4$$

Give the value of each of the following coefficients in the balanced equation.

$a =$
$b =$
$c =$
$d =$

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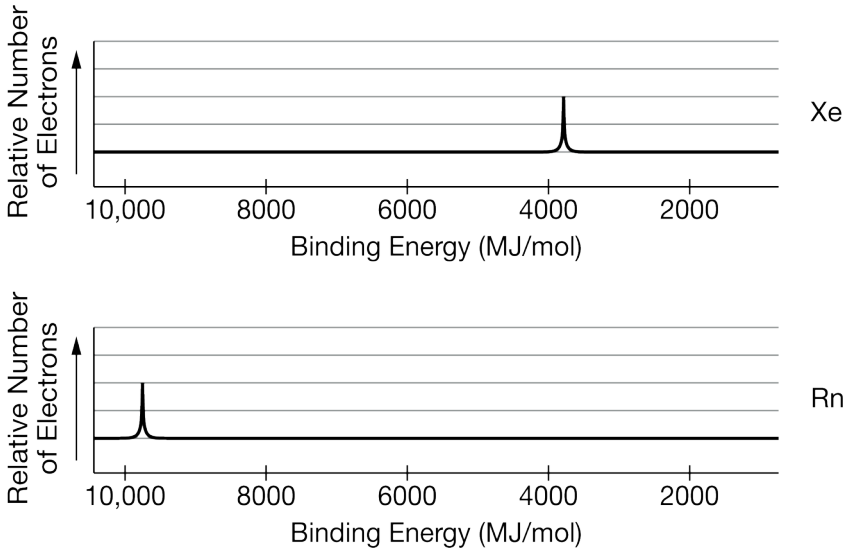
7. **Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

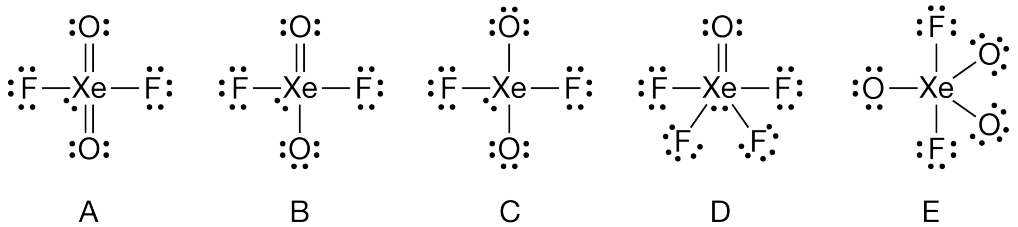
Answer the following questions about elements in Group 18 of the periodic table.

(a) Using principles of atomic structure, explain why the first ionization energy of Rn is less than that of Xe.

(b) Photoelectron spectroscopy data for the $1s$ sublevel of Xe and the $1s$ sublevel of Rn are represented below. In terms of Coulomb's law and atomic structure, explain why the peak for Xe is positioned so far to the right of the peak for Rn.



Three compounds that contain an element from Group 18 are $\mathrm{XeO_2F_2}$, $\mathrm{XeOF_4}$, and $\mathrm{XeO_3F_2}$. Possible Lewis electron-dot diagrams for the compounds are shown below.



(c) Which structure, A, B or C, minimizes the formal charges for each atom in $\mathrm{XeO_2F_2}$? Justify your choice.

(d) A student proposes that the energy of the $\mathrm{Xe-to-O}$ bond in the structure in diagram D is less than that of the $\mathrm{Xe-to-O}$ bond in the structure in diagram E. Do you agree or disagree with the student's claim? Justify your answer.

(e) The dipole moment of the $\mathrm{XeO_2F_2}$ molecule (diagrams A, B and C) is nonzero. Identify its molecular

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geometry and explain how its geometry and the polarity of its bonds result in a nonzero dipole moment.

The melting point of XeO_2F_2 (diagrams A, B, C) is greater than the melting point of XeO_3F_2 (diagram E).

(f) Identify the type(s) of intermolecular force(s) that the two substances have in common.

(g) Explain the difference in melting points based on the types and relative strengths of intermolecular forces.

(h) XeO_2F_2 reacts with SiO_2 to produce SiF_4 and XeO_3. The unbalanced chemical equation for the reaction appears below.

$$a\ XeO_2F_2 + b\ SiO_2 \rightarrow c\ SiF_4 + d\ XeO_3$$

Give the value of each of the following coefficients in the balanced equation.

$a =$
$b =$
$c =$
$d =$

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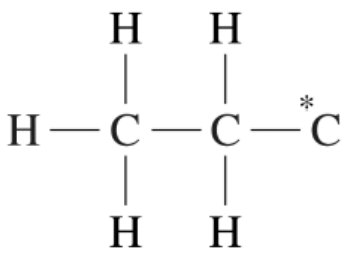
8. **Directions:** Examples and equations may be included in your responses where appropriate. For calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Pay attention to significant figures

Propanoic acid, C2H5COOH, is an organic acid that is a liquid at room temperature.

(a) An incomplete Lewis diagram for the propanoic acid molecule is provided in the box below. Complete the diagram, showing how the remaining atoms in the molecule are arranged around the carbon atom marked with an asterisk (*). Your structure should minimize formal charge and include any lone pairs of electrons.



(b) Identify the hybridization of the carbon atom marked with the asterisk

(c) Propanoic acid has a lower boiling point than butanoic acid, C_3H_7COOH.

i. Identify all the types of intermolecular forces present among the molecules in propanoic acid.

ii. Which of the types of intermolecular forces that you identified in part (c)(i) is most responsible for the difference in boiling points of the two acids?

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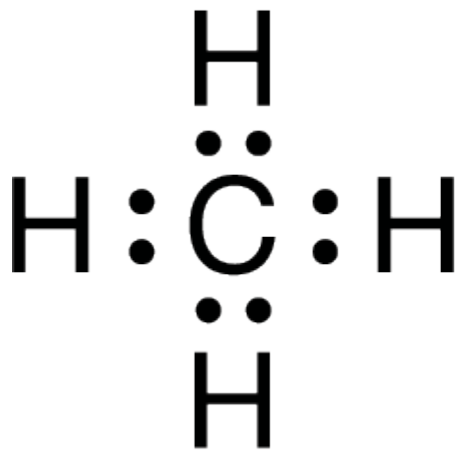
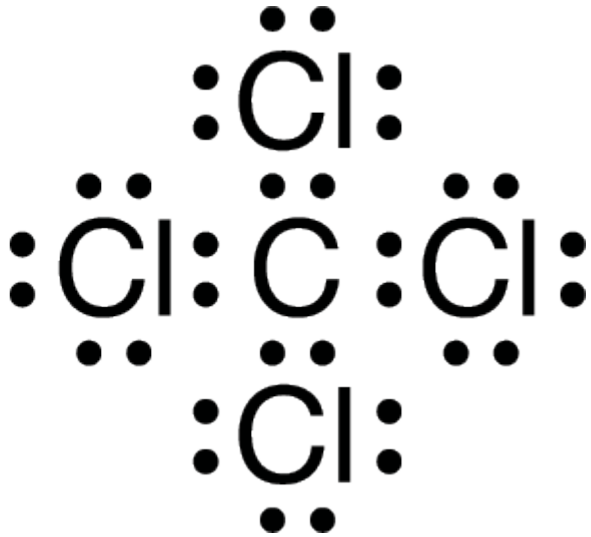
9. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.





Answer the following questions about CH_4 and CCl_4. The Lewis electron-dot diagrams for molecules of the compounds are shown above.

(a) Which compound has the higher boiling point? Justify your answer. The type(s) of intermolecular force(s) in both substances should be included in your answer.

(b) At $400\ \text{K}$ both compounds are gases. At this temperature, which compound, $CH_4(g)$ or $CCl_4(g)$, behaves more like an ideal gas? Justify your answer, including reasoning about both molecules.

(c) $CCl_4(l)$ is placed in a previously evacuated container at 30°C, and some of the $CCl_4(l)$ evaporates. In the box below, draw a particulate diagram to show the species in the container after some of the $CCl_4(l)$ has evaporated. The phases of the species should be indicated by the spacing and distribution of the particles in the diagram. Some of the species shown in the legend will not be used.

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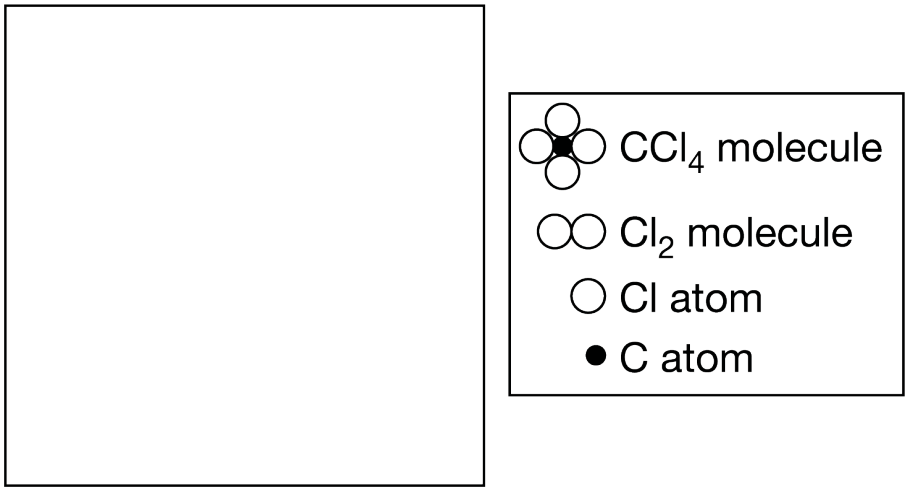
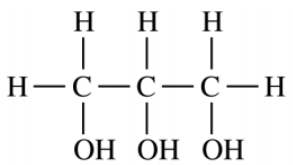
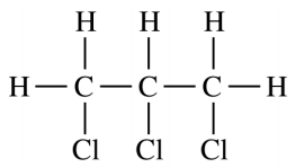


10.



Glycerol

Boiling point 290°C



Trichloropropane

Boiling point 157°C

The structural formulas of glycerol and trichloropropane are given above. Both compounds are liquids at 25°C.

a. For each compound, identify all types of intermolecular forces present in the liquid. Explain why glycerol has the higher boiling point in terms of the relative strengths of the intermolecular forces.

b. Glycerol (molar mass 92.09 g/mol) has been suggested for use as an alternative fuel. The enthalpy of combustion, $\Delta H°_{comb}$, of glycerol is -1654 kJ/mol. What mass of glycerol would need to be combusted to heat 500.0 g of water from 20.0°C to 100.0°C ? (The specific heat capacity of water is 4.184 J/(g·°C). Assume that all the heat released by the combustion reaction is absorbed by the water.)

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11. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions related to $\mathrm{HBr}(l)$ and $\mathrm{HF}(l)$.

(a) In the following table, list all of the types of intermolecular forces present in pure samples of $\mathrm{HBr}(l)$ and $\mathrm{HF}(l)$.

Liquid	$\mathrm{HBr}(l)$	$\mathrm{HF}(l)$
Intermolecular forces present		

(b) The enthalpy of vaporization, ΔH°_{vap}, for each liquid is provided in the following table.

Liquid	$\mathrm{HBr}(l)$	$\mathrm{HF}(l)$
ΔH°_{vap}	$17.3\ \mathrm{kJ/mol}$	$25.2\ \mathrm{kJ/mol}$

(i) Based on the types and relative strengths of intermolecular forces, explain why ΔH°_{vap} of $\mathrm{HF}(l)$ is greater than that of $\mathrm{HBr}(l)$.

(ii) Calculate the amount of thermal energy, in kJ, required to vaporize $6.85\ \mathrm{g}$ of $\mathrm{HF}(l)$.

(c) Based on the arrangement of electrons in the Br and F atoms, explain why the bond length in an HBr molecule is greater than that in an HF molecule.

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12.

Compound	Melting Point (°C)
LiI	449
KI	686
LiF	845
NaF	993

A student learns that ionic compounds have significant covalent character when a cation has a polarizing effect on a large anion. As a result, the student hypothesizes that salts composed of small cations and large anions should have relatively low melting points.

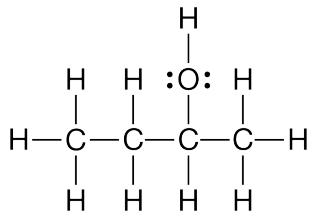
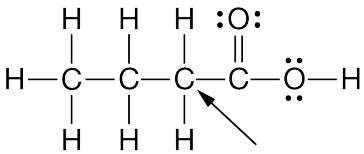
a. Select two compounds from the table and explain how the data support the student's hypothesis.

b. Identify a compound from the table that can be dissolved in water to produce a basic solution. Write the net ionic equation for the reaction that occurs to cause the solution to be basic.

frq6

13. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student investigates the physical and chemical properties of various carbon-containing compounds. The complete Lewis electron-dot diagrams and boiling points for two compounds, L and X, are shown in the following table.

Compound	L	X
Diagram		
Boiling Point	371 K	437 K

(a) Identify the hybridization of the valence orbitals of the carbon atom in compound X that is indicated by the arrow in the diagram.

(b) The $\mathrm{C-O}$ bond in compound L is longer than any $\mathrm{C-H}$ bond in compound X. Explain the reason for this difference using principles of atomic structure.

(c) For each compound, list all intermolecular forces present.

Compound L –
Compound X –

(d) Explain why the boiling point of compound X is higher than that of compound L based on the types and relative strengths of the intermolecular forces present in each substance.

(e) A $0.10\ \mathrm{mol}$ sample of compound L is placed in an evacuated, rigid container at $400.\ \mathrm{K}$ and $0.25\ \mathrm{atm}$. When more moles of compound L are added, the pressure increases to $0.75\ \mathrm{atm}$ at $400.\ \mathrm{K}$. Determine the number of moles of compound L now present in the container.

Another carbon-containing compound is $\mathrm{C_2H_6O}$, which reacts with $\mathrm{H_2CrO_4}$, as represented by the following equation.

$$\mathrm{C_2H_6O + H_2CrO_4 \rightarrow C_2H_4O + H_2CrO_3 + H_2O}$$

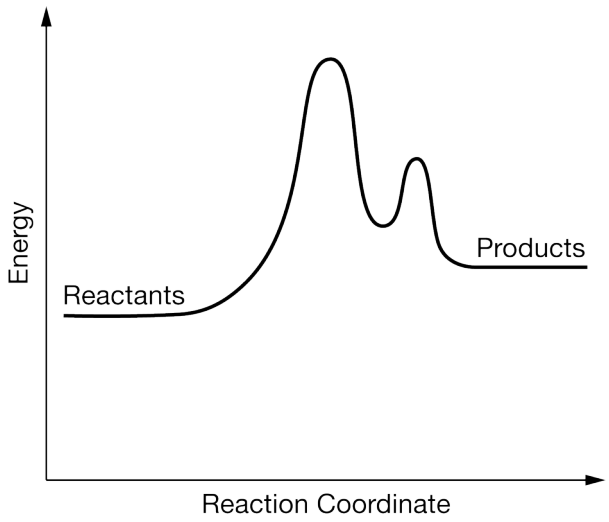
(f) The student claims that C is oxidized when $\mathrm{C_2H_6O}$ is converted to $\mathrm{C_2H_4O}$. Do you agree or disagree? Justify your answer in terms of oxidation numbers.

frq6

The reaction is known to be exothermic. A proposed mechanism for the reaction is shown in the following table.

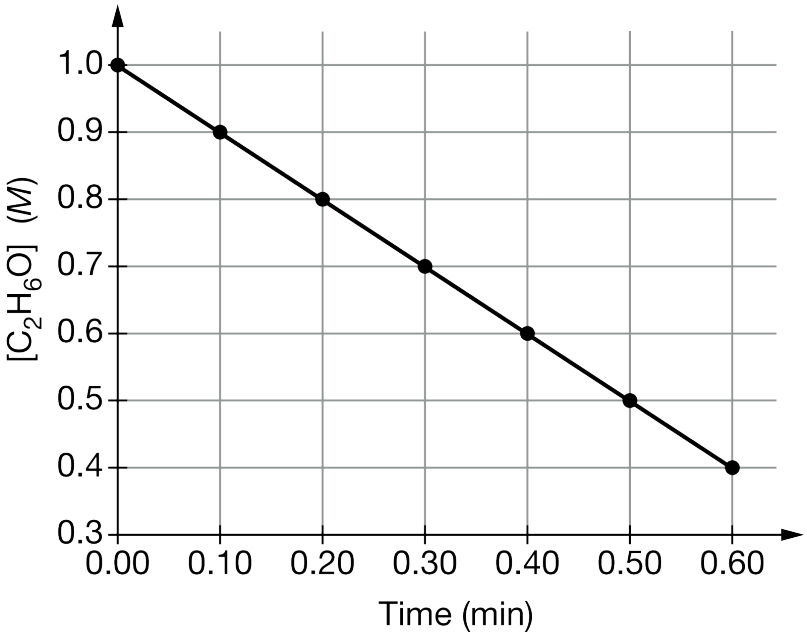
Step 1	$C_2H_6O + H_2CrO_4 \rightleftarrows C_2H_5CrO_4H + H_2O$	*fast*
Step 2	$C_2H_5CrO_4H \rightarrow C_2H_4O + H_2CrO_3$	*slow*

(g) The student draws the following energy diagram for the reaction. Is the drawing consistent with the proposed mechanism? Justify your answer.



(h) The reaction is run in an excess of H_2CrO_4 such that $[H_2CrO_4]$ remains constant. The student claims that under these conditions, the reaction is zeroth-order with respect to C_2H_6O, and that the rate constant of the reaction is $1.0\ M/\text{min}$. Is the following graph consistent with the student's claim? Justify your answer with respect to the graph's axes and the numerical value of the slope of the line.

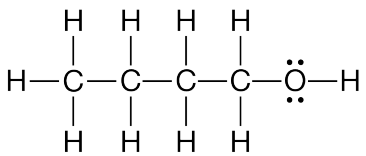
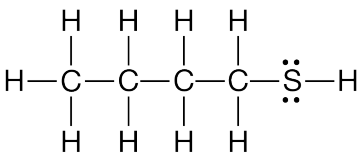
frq6


1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
[C2H6O] (M)
0.00
0.10
0.20
0.30
0.40
0.50
0.60
Time (min)

frq6

14. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student investigates the physical and chemical properties of various carbon-containing compounds. The complete Lewis electron-dot diagrams and boiling points for two compounds, L and Z, are shown in the following table.

Compound	L	Z
Diagram		
Boiling Point	391 K	371 K

(a) Based on VSEPR theory, what is the approximate value of the $\mathrm{C-O-H}$ bond angle in compound L?

(b) The $\mathrm{C-O}$ bond in compound L is shorter than the $\mathrm{C-S}$ bond in compound Z. Explain the reason for this difference using principles of atomic structure.

(c) For each compound, list all intermolecular forces present.

Compound L –
Compound Z –

(d) Explain why the boiling point of compound L is higher than that of compound Z based on the types and relative strengths of the intermolecular forces present in each substance.

(e) A gaseous sample of compound Z is placed in an evacuated, rigid container at a temperature of $400.\ \mathrm{K}$ and a pressure of $0.50\ \mathrm{atm}$. Determine the final pressure when the temperature of the sample is increased to $600.\ \mathrm{K}$.

Another carbon-containing compound is $\mathrm{C_2H_6O}$, which reacts with $\mathrm{H_2CrO_4}$, as represented by the following equation.

$$\mathrm{C_2H_6O + H_2CrO_4 \rightarrow C_2H_4O + H_2CrO_3 + H_2O}$$

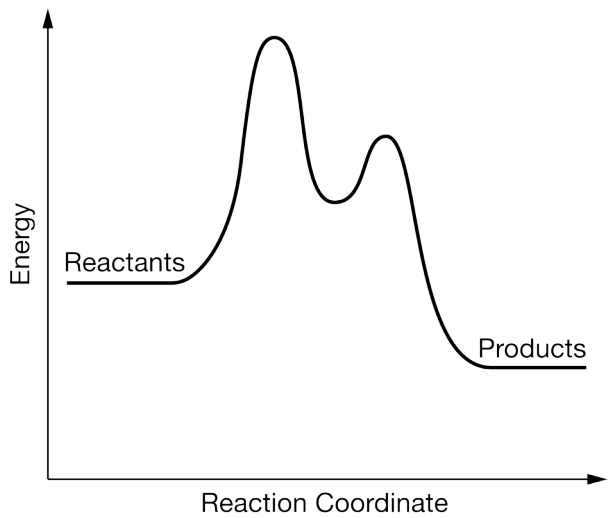
(f) The student claims that O is oxidized when $\mathrm{C_2H_6O}$ is converted to $\mathrm{C_2H_4O}$. Do you agree or disagree? Justify your answer in terms of oxidation numbers.

The reaction is known to be exothermic. A proposed mechanism for the reaction is shown in the following table.

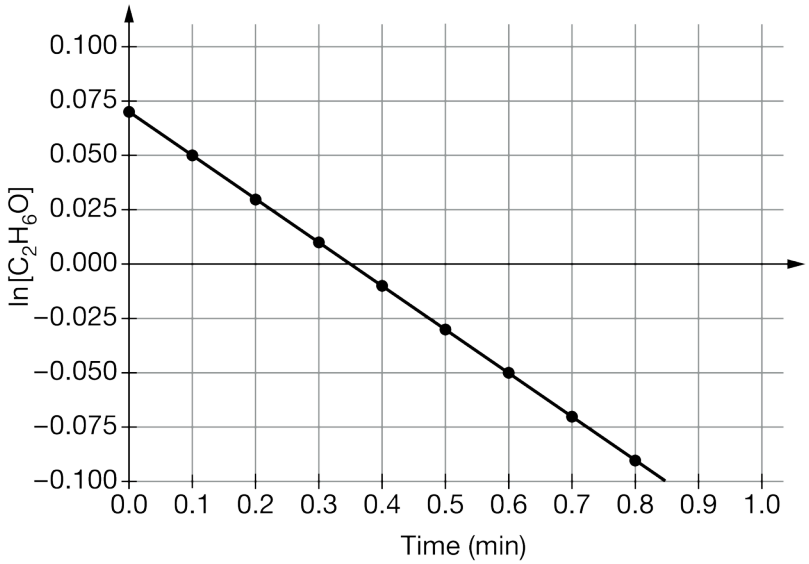
frq6

Step 1	$C_2H_6O + H_2CrO_4 \rightleftarrows C_2H_5CrO_4H + H_2O$	*fast*
Step 2	$C_2H_5CrO_4H \rightarrow C_2H_4O + H_2CrO_3$	*slow*

(g) The student draws the following energy diagram for the reaction. Is the drawing consistent with the proposed mechanism? Justify your answer.



(h) The reaction is run in an excess of H_2CrO_4 such that $[H_2CrO_4]$ remains constant. The student claims that under these conditions, the reaction is first-order with respect to C_2H_6O, and that the rate constant of the reaction is $0.50\ \text{min}^{-1}$. Is the following graph consistent with the student's claim? Justify your answer with respect to the graph's axes and the numerical value of the slope of the line.



frq6

15. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

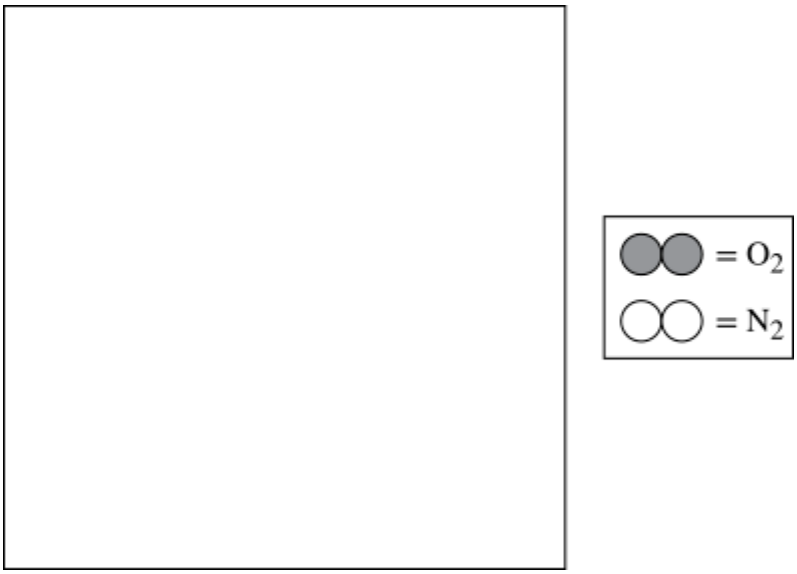
Dry air is comprised of N_2, O_2, and traces of other gases.

A student obtains a $2.00\ L$ container filled with a sample of dry air at a total pressure of $1.30\ atm$ and a temperature of $295\ K$.

(a) Calculate the total number of moles of gas in the container.

(b) The student cools the gaseous mixture until either N_2 or O_2 condenses. Which gas will condense first? Justify your answer in terms of the types and relative strengths of the intermolecular forces in N_2 and O_2.

(c) Using the legend given, draw at least three molecules of each type of gas in the sealed container represented below when one of the substances has condensed, but the other substance is still a gas.



(d) In this sample, 78% of the molecules are N_2 and 21% of the molecules are O_2. Calculate the total number of grams of the substance that condenses first.

frq6

16. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about the compounds NH_2Cl and NCl_3. The Lewis electron-dot diagrams of the two compounds are shown.

(a) Calculate the number of moles of NH_2Cl (molar mass $51.48\ \text{g/mol}$) present in $1.0\ \text{L}$ of a solution in which the concentration of NH_2Cl is $0.0016\ \text{g/L}$.

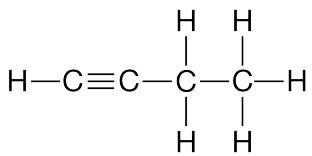
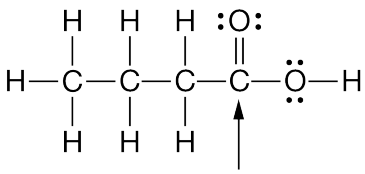
(b) NH_2Cl is highly soluble in water, whereas NCl_3 is nearly insoluble. Explain this observation in terms of the types and relative strengths of the intermolecular forces between each of the solutes and water.

(c) The value of $\Delta H^{\circ}_{vaporization}$ for $NCl_3(l)$ is $32.9\ \text{kJ/mol}$. Calculate the amount of energy required to vaporize a $15.0\ \text{g}$ sample of NCl_3 (molar mass $120.36\ \text{g/mol}$).

frq6

17. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student investigates the physical and chemical properties of various carbon-containing compounds. The complete Lewis electron-dot diagrams and boiling points for two compounds, Q and X, are shown in the following table.

Compound	Q	X
Diagram		
Boiling Point	281 K	437 K

(a) Identify the hybridization of the valence orbitals of the carbon atom in compound X that is indicated by the arrow in the diagram.

(b) The $\mathrm{C-H}$ bonds in compound Q are shorter than the $\mathrm{C-C}$ bonds in compound X. Explain the reason for this difference using principles of atomic structure.

(c) For each compound, list all intermolecular forces present.

Compound Q –
Compound X –

(d) Explain why the boiling point of compound X is higher than that of compound Q based on the types and relative strengths of the intermolecular forces present in each substance.

(e) A gaseous sample of compound Q is placed in an evacuated cylinder with a movable piston that can change the volume of the cylinder. The initial volume and temperature of the gas are $10.0\ \mathrm{L}$ and $400.\ \mathrm{K}$. If the temperature of the gas is raised to $500.\ \mathrm{K}$ as its pressure remains constant, what is the final volume of the gas?

Another carbon-containing compound is $\mathrm{C_2H_6O}$, which reacts with $\mathrm{H_2CrO_4}$, as represented by the following equation.

$$\mathrm{C_2H_6O + H_2CrO_4 \rightarrow C_2H_4O + H_2CrO_3 + H_2O}$$

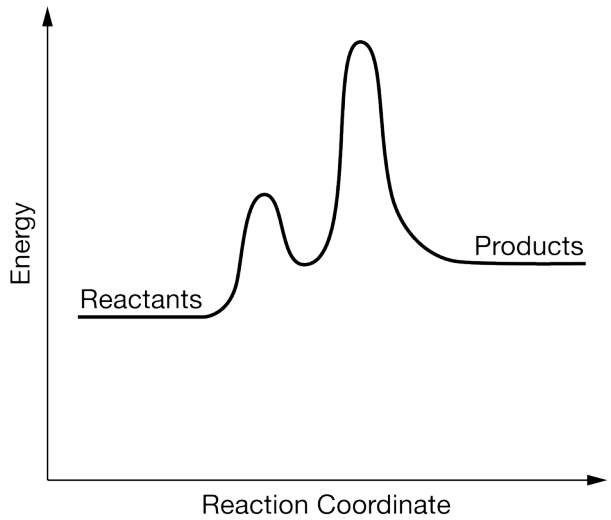
(f) The student claims that Cr is reduced when $\mathrm{H_2CrO_4}$ is converted to $\mathrm{H_2CrO_3}$. Do you agree or disagree? Justify your answer in terms of oxidation numbers.

frq6

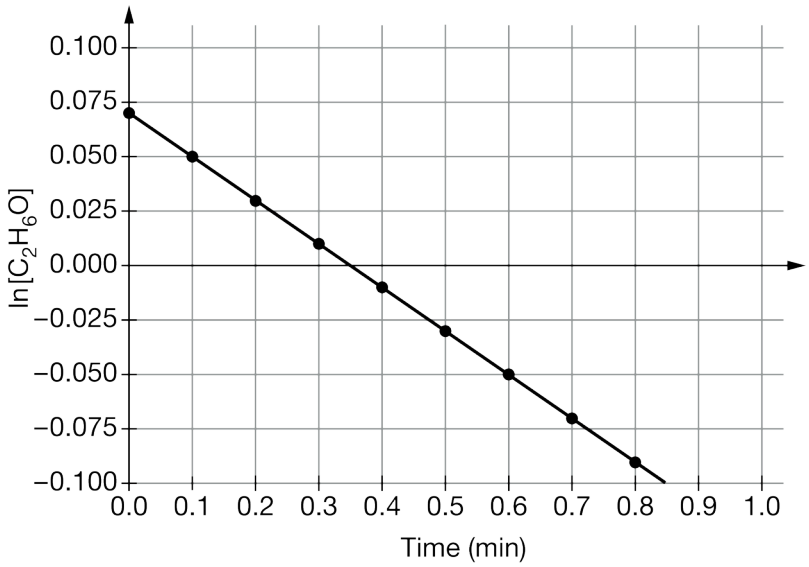
The reaction is known to be exothermic. A proposed mechanism for the reaction is shown in the following table.

Step 1	$C_2H_6O + H_2CrO_4 \rightleftarrows C_2H_5CrO_4H + H_2O$	*fast*
Step 2	$C_2H_5CrO_4H \rightarrow C_2H_4O + H_2CrO_3$	*slow*

(g) The student draws the following energy diagram for the reaction. Is the drawing consistent with the proposed mechanism? Justify your answer.



(h) The reaction is run in an excess of H_2CrO_4 such that $[H_2CrO_4]$ remains constant. The student claims that under these conditions, the reaction is second-order with respect to C_2H_6O, and that the rate constant of the reaction is $0.20\ M^{-1}\ \text{min}^{-1}$. Is the following graph consistent with the student's claim? Justify your answer with respect to the graph's axes and the numerical value of the slope of the line.



frq6

18. For parts of the free response question that require calculations, clearly show the method and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

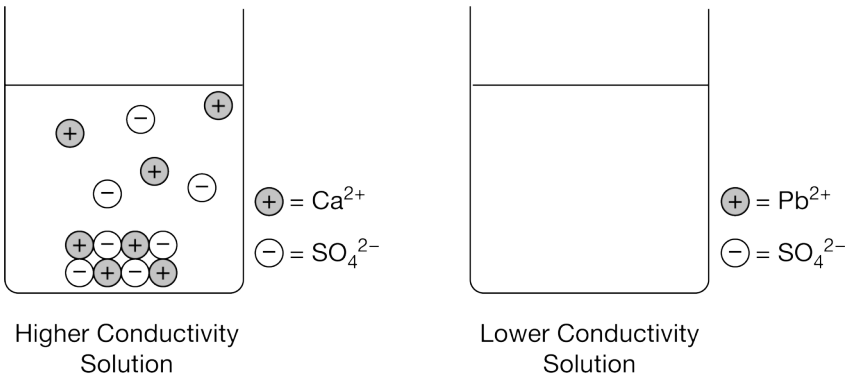
A student is studying the properties of $CaSO_4$ and $PbSO_4$. The student has samples of both compounds, which are white powders.

(a) The student tests the electrical conductivity of each solid and observes that neither solid conducts electricity. Describe the structures of the solids that account for their inability to conduct electricity.

The student places excess $CaSO_4(s)$ in a beaker containing $100\ mL$ of water and places excess $PbSO_4(s)$ in another beaker containing $100\ mL$ of water. The student stirs the contents of the beakers and then measures the electrical conductivity of the solution in each beaker. The student observes that the conductivity of the solution in the beaker containing the $CaSO_4(s)$is higher than the conductivity of the solution in the beaker containing the $PbSO_4(s)$.

(b) Which compound is more soluble in water, $CaSO_4(s)$ or $PbSO_4(s)$? Justify your answer based on the results of the conductivity test.

The left side of the diagram below shows a particulate representation of the contents of the beaker containing the $CaSO_4(s)$ from the solution conductivity experiment.



(c) Draw a particulate representation of $PbSO_4(s)$ and the ions dissolved in the solution in the beaker on the right in the diagram. Draw the particles to look like those shown to the right of the beaker. Draw an appropriate number of dissolved ions relative to the number of dissolved ions in the beaker on the left.

(d) The student attempts to increase the solubility of $CaSO_4(s)$ by adding $10.0\ mL$ of $2\ M\ H_2SO_4(aq)$ to the beaker, and observes that additional precipitate forms in the beaker. Explain this observation.

frq6

19. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Some building materials contain small capsules filled with paraffin wax to improve the insulating properties of the materials. The melting and freezing of paraffin inside the capsules in the material helps to regulate temperature. Paraffin wax begins to melt at $37^{\circ}\mathrm{C}$.

(a) While the wax is melting, is the net flow of thermal energy from the wax to the surroundings or from the surroundings to the wax? Justify your answer.

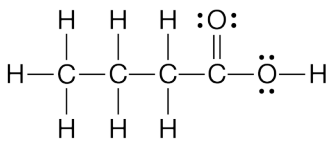
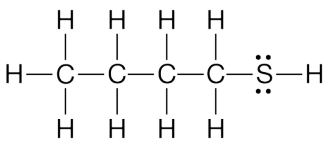
(b) Calculate the amount of thermal energy, in kJ, that is required to melt 15.2 grams of solid paraffin wax when the temperature of the surroundings is above the melting point of paraffin. (The molar mass of paraffin is $282.62\ \mathrm{g/mol}$, and its molar heat of fusion is $48.78\ \mathrm{kJ/mol}$.)

(c) When paraffin wax melts, its density decreases. Provide a particle-level explanation of this observation.

frq6

20. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student investigates the physical and chemical properties of various carbon-containing compounds. The complete Lewis electron-dot diagrams and boiling points for two compounds, X and Z, are shown in the following table.

Compound	X	Z
Diagram		
Boiling Point	437 K	371 K

(a) Based on VSEPR theory, what is the approximate value of the $\text{O} - \text{C} - \text{O}$ bond angle in compound X?

b) The $\text{O} - \text{H}$ bond in compound X is shorter than the $\text{S} - \text{H}$ bond in compound Z. Explain the reason for this difference using principles of atomic structure.

(c) For each compound, list all intermolecular forces present.

Compound X –
Compound Z –

(d) Explain why the boiling point of compound X is higher than that of compound Z based on the types and relative strengths of the intermolecular forces present in each substance.

(e) A gaseous sample of compound Z is placed in an evacuated, rigid container at a temperature of $800.\ \text{K}$ and a pressure of $1.00\ \text{atm}$. Determine the final pressure of the gas when the temperature is decreased to $600.\ \text{K}$.

Another carbon-containing compound is C_2H_6O, which reacts with H_2CrO_4, as represented by the following equation.

$$C_2H_6O + H_2CrO_4 \rightarrow C_2H_4O + H_2CrO_3 + H_2O$$

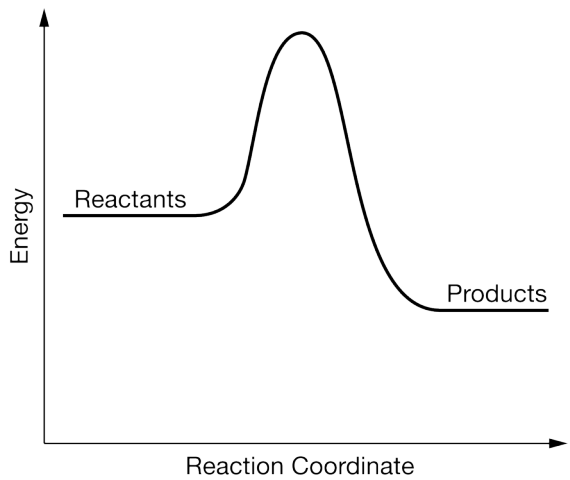
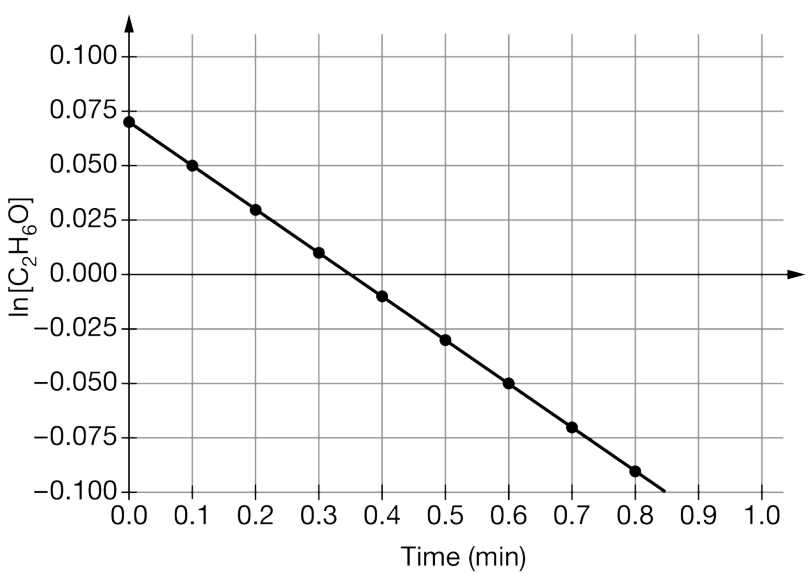
(f) The student claims that C is reduced when C_2H_6O is converted to C_2H_4O. Do you agree or disagree? Justify your answer in terms of oxidation numbers.

The reaction is known to be exothermic. A proposed mechanism for the reaction is shown in the following table.

frq6

Step 1	$C_2H_6O + H_2CrO_4 \rightleftarrows C_2H_5CrO_4H + H_2O$	*fast*
Step 2	$C_2H_5CrO_4H \rightarrow C_2H_4O + H_2CrO_3$	*slow*

(g) The student draws the following energy diagram for the reaction. Is the drawing consistent with the proposed mechanism? Justify your answer.



(h) The reaction is run in an excess of H_2CrO_4 such that $[H_2CrO_4]$ remains constant. The student claims that under these conditions, the reaction is zeroth-order with respect to C_2H_6O, and that the rate constant of the reaction is $0.20\ M/\text{min}$. Is the following graph consistent with the student's claim? Justify your answer with respect to the graph's axes and the numerical value of the slope of the line.